

## AI Use Declaration Form

**Course:** Introduction to Artificial Intelligence

**Lab Title:** Lab 4: LLMs and Prompt Engineering for Decision Support

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**GitHub Repository Link:** <https://github.com/Rems-dev1/lab-4-llm-decision-support>

**Date Submitted:** 15/08/2026

### 1. AI Use Summary

| Question                                            | Student Response |
|-----------------------------------------------------|------------------|
| Did you use any AI tool for this lab?               | Yes              |
| Estimated percentage of the work influenced by AI   | 17%              |
| Did you attach evidence of AI use where applicable? | Yes              |

### 2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

| Name of AI Tool Used | Purpose for Using the Tool | Prompt or Instruction Given to the Tool | Part(s) of the Work Influenced by the Tool | How I Verified, Edited, or Improved the AI Output |
|----------------------|----------------------------|-----------------------------------------|--------------------------------------------|---------------------------------------------------|
|                      |                            |                                         |                                            |                                                   |

|        |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                  |                                                                                                          |
|--------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Gemini | Explaining concepts and guidance | <p>I want you to act as my personal AI, Python, and Algorithms tutor. I am a complete beginner. Assume I have never studied Artificial Intelligence.,.</p> <p>Your goal is to help me fully understand the attached lab document so that I can complete every task myself and confidently explain it to someone else.</p> <p>Important teaching rules:</p> <p>Do NOT directly answer any TODOs, coding exercises, self-checks, reflection questions, or assignment questions unless I explicitly ask for the answer later.</p> <p>Teach me step-by-step before expecting me to write code.</p> | Explaining to me the meaning of the output and interpreting things which are challenging for me. | I verified to see if the output does not have errors or is out of range of the scope of these exercises. |
|        |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                  |                                                                                                          |
|        |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                  |                                                                                                          |

### 3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

| Evidence Type                         | Attached? (Yes, No, N/A) | File Name                    |
|---------------------------------------|--------------------------|------------------------------|
| AI-generated draft                    | No                       |                              |
| Prompt history                        | No                       |                              |
| Screenshot of AI interaction          | Yes                      | Found at bottom of this page |
| Exported AI conversation              | No                       |                              |
| AI-generated code snippet             | NO                       |                              |
| Revised version showing student input | No                       |                              |

### 4. Student Declaration

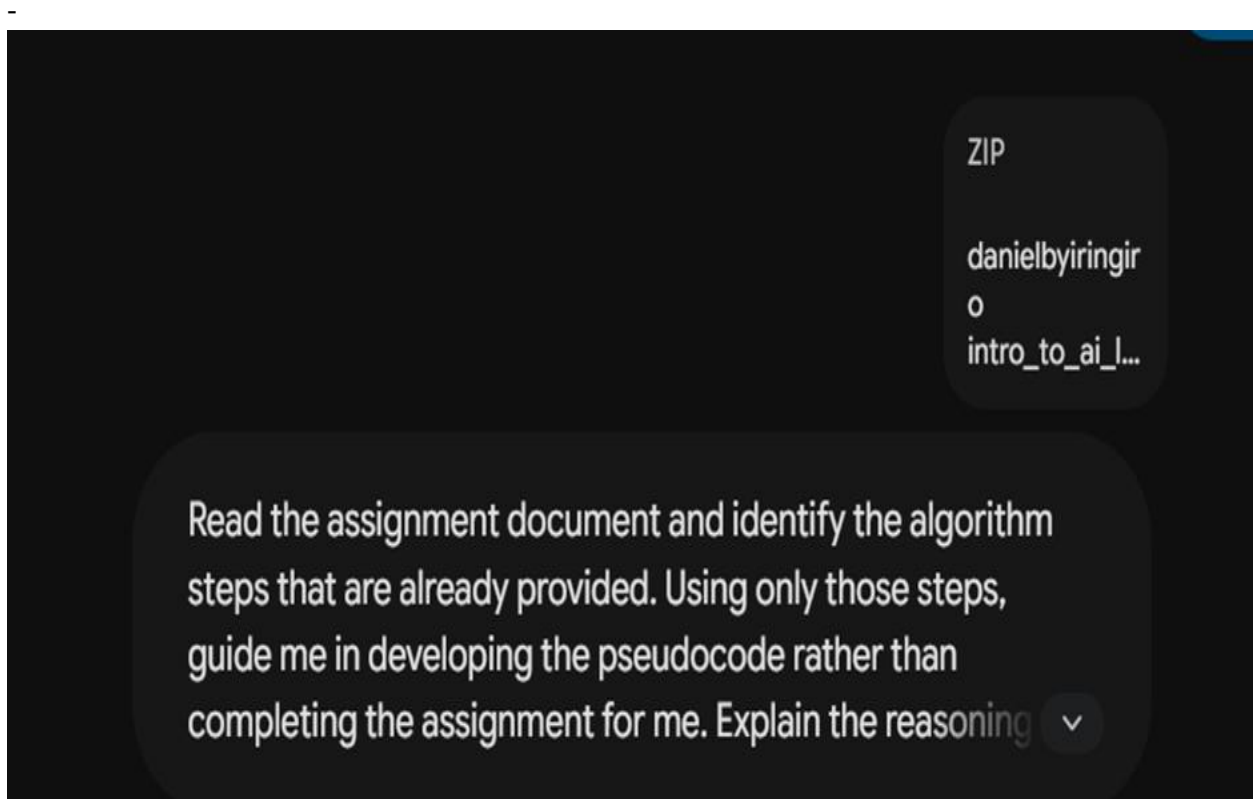
I declare that:

| Declaration Statement                                                                     | Response (Yes/No) |
|-------------------------------------------------------------------------------------------|-------------------|
| The submitted work is my own work.                                                        | Yes               |
| Any use of AI tools has been clearly declared in this form.                               | Yes               |
| The prompts, instructions, and AI-generated outputs have been disclosed where applicable. | Yes               |
| I reviewed, tested, edited, and improved any AI-generated content before submission.      | Yes               |
| My AI usage does not exceed 25% of the entire work.                                       | Yes               |
| I understand that undeclared or excessive AI use may be treated as academic misconduct.   | Yes               |

**Student Signature:** Remy Nteziryayo

**Date:** 15/08/2026

Attachments:



AI answer:

Plaintext



```
FUNCTION ask_llm(user_prompt, system_prompt, temperature, max_tokens):  
    CREATE completion request using client WITH:  
        - model = MODEL  
        - messages = [system role message, user role message]  
        - temperature = temperature  
        - max_tokens = max_tokens  
  
    PRINT total tokens used from response usage data  
    RETURN message content from response choice  
  
CALL ask_llm WITH test question  
PRINT output answer
```

## Part 1.2: Temperature Experiment

IPYNB

Plaintext



```
SET test_prompt = "Suggest a single name for a savings product..."  
  
LOOP 5 times:  
    CALL ask_llm(test_prompt, temperature = 0.0)  
    PRINT result  
  
LOOP 5 times:  
    CALL ask_llm(test_prompt, temperature = 1.2)  
    PRINT result
```

## Part 3.1: Summarization Component

IPYNB

Plaintext



```
# Naive Prompt (V1)
SET SUMMARY_PROMPT_V1 = "Summarize this:"
CALL ask_llm WITH V1 ON letter L002 AND L006
PRINT responses

# Structured Prompt (V2)
SET SYSTEM_PROMPT_V2 = "Role: Microfinance assistant. Output: Factual 3-4 sentence summary"
SET USER_TEMPLATE = "Summarize this loan application:\n\n{letter_text}"

CALL ask_llm WITH USER_TEMPLATE ON letter L002 AND L006 AT temperature = 0.0
PRINT responses
```

## Part 3.2: Structured Extraction (JSON)

IPYNB

Plaintext



```
SET EXTRACT_SYSTEM_PROMPT = ""
Target Output: JSON object with keys (applicant_name, amount_ghs, purpose, monthly_payment)
Rule: If missing, return null. Do not guess.
Few-Shot Example: [Provide 1 mock letter and 1 mock JSON output]
""

FUNCTION extract_fields(letter_text):
    CALL ask_llm(letter_text, system_prompt = EXTRACT_SYSTEM_PROMPT, temperature = 0.0)
```

+ Ask Gemini



Extended

